

Methodology

AI-Powered Reputation Intelligence System

AI & Data Solutions Specialist Assignment

1. Introduction

The objective of this project is to develop an AI-powered Reputation Intelligence System capable of transforming unstructured digital media mentions into structured business insights. The solution processes a dataset of news and digital articles related to ICICI Prudential AMC, classifies each mention into predefined Reputation Drivers and Sub-drivers using a Large Language Model (LLM), and presents the results through an interactive Streamlit dashboard.

The overall workflow consists of five major stages:

1. Data Exploration
 2. Data Preprocessing
 3. AI-based Classification
 4. Insight Generation
 5. Interactive Dashboard Development
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2. Data Exploration

The provided dataset was initially analyzed to understand its structure and quality.

The exploratory analysis included:

- Number of records and attributes
- Missing value identification
- Duplicate record detection
- Data type validation
- Text field examination

The primary textual attributes used for downstream processing were:

- Title
- Opening Text
- Hit Sentence

These fields were merged to create a unified text representation for each article.

3. Data Preprocessing

The preprocessing stage focused on improving data quality before AI classification.

The following operations were performed:

Duplicate Removal

Duplicate records were identified and removed to avoid redundant classification.

Missing Value Handling

Missing values in textual fields were replaced with empty strings to ensure consistent text concatenation.

Sentiment Standardization

Existing sentiment labels were normalized into a consistent format (Positive, Neutral, Negative).

Text Consolidation

The following fields were merged into a single feature named **combined_text**:

- Title
- Opening Text
- Hit Sentence

This combined representation provided sufficient context for the AI model.

Source Standardization

Source names were standardized to improve readability and reporting.

4. AI Classification Methodology

The core intelligence layer uses the **Mistral AI API** to classify each media mention.

Prompt engineering was employed to constrain the model to a predefined taxonomy, ensuring consistent outputs.

Each article is classified into:

Reputation Drivers

- Brand Perception
- User Experience
- Responsible Business Practices

Each Driver is mapped to one corresponding Sub-driver.

Brand Perception

- Thought Leadership
- Product Strategy
- Brand Visibility & Marketing

User Experience

- Product & Service Quality
- Customer Support & Complaint Resolution
- Digital & Omnichannel Experience

Responsible Business Practices

- Regulatory Compliance & Ethical Governance
- Social Impact & Community (CSR)

The model returns structured JSON, which is validated before being stored in the classified dataset.

5. Insight Generation

After classification, analytical insights were generated using Python.

The following analyses were performed:

- Total digital mentions
- Sentiment distribution
- Reputation Driver distribution
- Sub-driver distribution
- Positive reputation drivers
- Negative reputation drivers
- Top discussion themes
- Word cloud generation

These insights provide an overview of the organization's digital reputation.

6. Dashboard Development

A Streamlit-based dashboard was developed to visualize the analytical results.

The dashboard consists of three major modules:

Overview

- KPI Cards
- Sentiment Distribution
- Driver Distribution
- Sub-driver Distribution
- Word Cloud
- Top Discussion Themes

Content Explorer

Allows users to:

- Search articles
- Filter by Driver
- Filter by Sub-driver
- Filter by Sentiment
- View original content
- Download filtered datasets

AI Insights

Displays:

- Positive reputation drivers
 - Negative reputation drivers
 - Executive summary
 - Business recommendations
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7. Assumptions

The following assumptions were made during implementation:

- Existing sentiment labels were considered accurate.
 - Every article belongs to one primary Reputation Driver.
 - AI responses follow the predefined taxonomy.
 - The provided dataset is representative of digital reputation coverage.
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8. Limitations

The current implementation has several limitations:

- Static dataset
 - No real-time data ingestion
 - English-language content only
 - AI classification depends on prompt quality
 - External API availability affects execution
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9. Conclusion

The developed system successfully transforms unstructured reputation data into structured business intelligence through AI-powered classification and interactive visualization.

The modular architecture enables future integration with real-time data sources, automated workflows, and enterprise-scale analytics platforms while maintaining flexibility and scalability.